

Predicting NYC Household Income

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```
library("knitr")
library("cmu202")
library("kableExtra")
library("pander")
library("readr")
library("magrittr")
library("car")
library("jtools")
library("leaps")
```

```
nyc <- readr::read_csv("http://stat.cmu.edu/~gordonw/nyc.csv")
```

Introduction

New York City is known as the largest city in the United States and contains some of the country's most well-known celebrities and attractions. Many people aspire to visit New York City sometime during their lifetimes, and some even wish to live there someday. However, housing prices in this city far exceed other parts of the United States, and it takes a certain minimum level of income in order to survive in New York City.

In order to figure out the minimum income needed to live in New York City, this report aims to predict the annual income of residents in the city. We have pulled data from the New York City Housing and Vacancy Survey in order to get an idea of a typical New York City resident's annual income.

Exploratory Data Analysis

Data

Data is taken from The New York City Housing and Vacancy Survey, a survey designed to accurately understand the housing conditions in New York City. This survey is conducted every three years, and has an admirably high response rate.

For this dataset, we analyze a random sample of 299 respondents and 4 variables.

The variables are as follows:

- *Income*: Total household income, in dollars
- *Age*: The respondent's age, in years
- *MaintenanceDef*: Number of maintenance deficiencies of the residence, between 2002 and 2005
- *NYCMove*: The year the respondent moved to New York City

The first few lines of the data appear as follows:

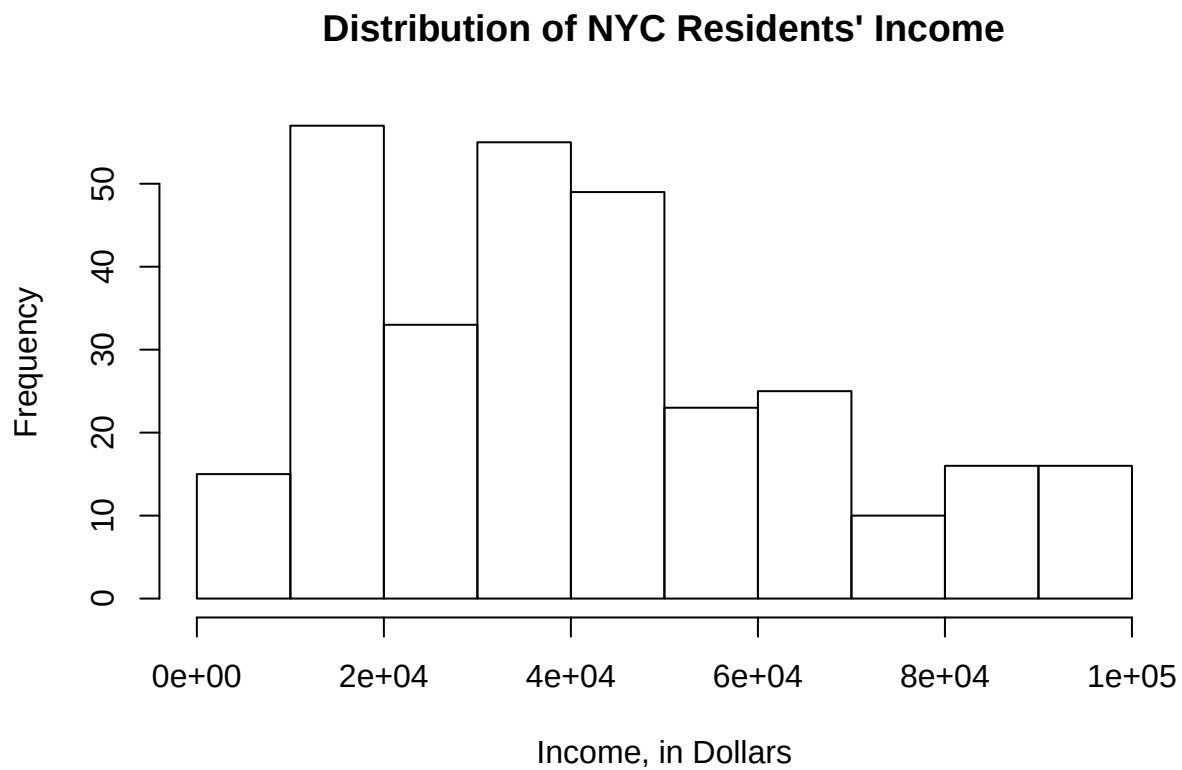
```
## # A tibble: 6 x 4
##   Income   Age MaintenanceDef NYCMove
##   <dbl> <dbl>          <dbl>   <dbl>
## 1   8400    77             1    1981
## 2  17510    53             2    1986
## 3  19200    33             4    1992
## 4  42717    55             1    1969
## 5   5000    58             2    1989
## 6  30000    29             4    1994
```

Univariate Exploration

To understand the data we are working with, we will first explore each of the individual variables in our dataset.

Below are histogram plots to display the distributions of Income, Age, MaintenanceDef, and NYCMove.

Income



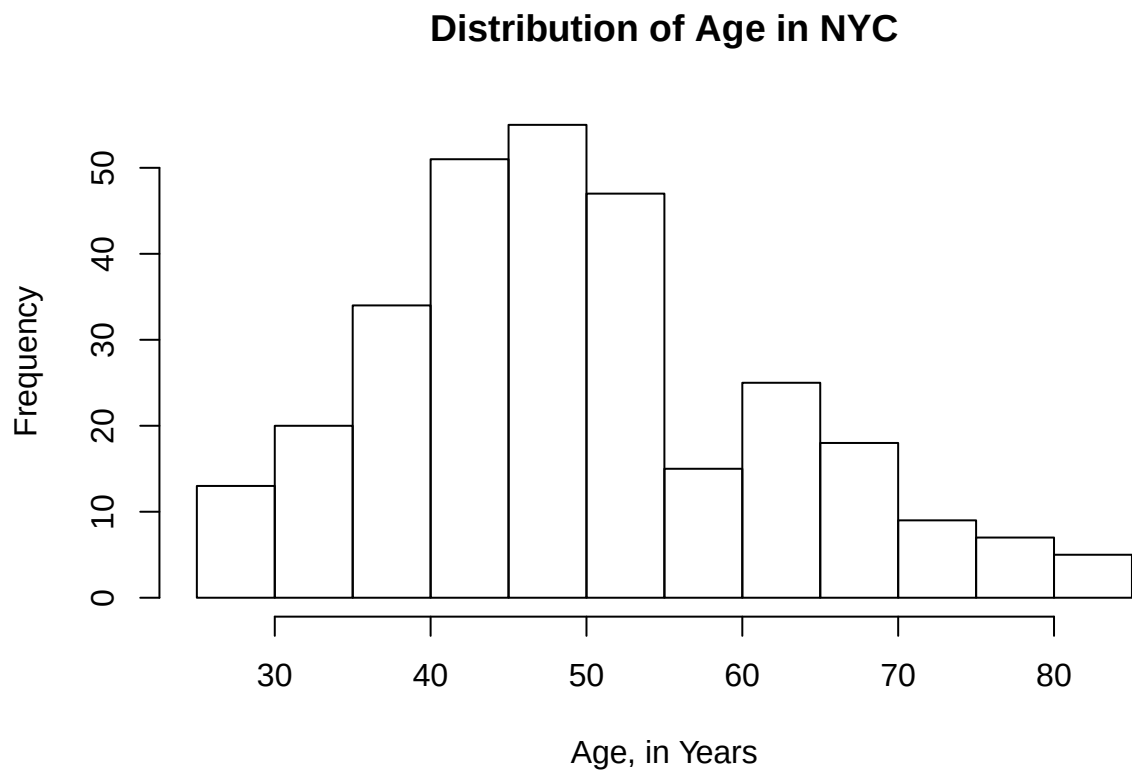
A numerical summary is supplemented as follows:

```
summary(nyc$Income)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	1440	21000	39000	42266	57800	98000

We notice that the distribution of *income* in NYC is approximately unimodal and right-skewed, with an average of \$42,266. There are no significant outliers in the data.

Age

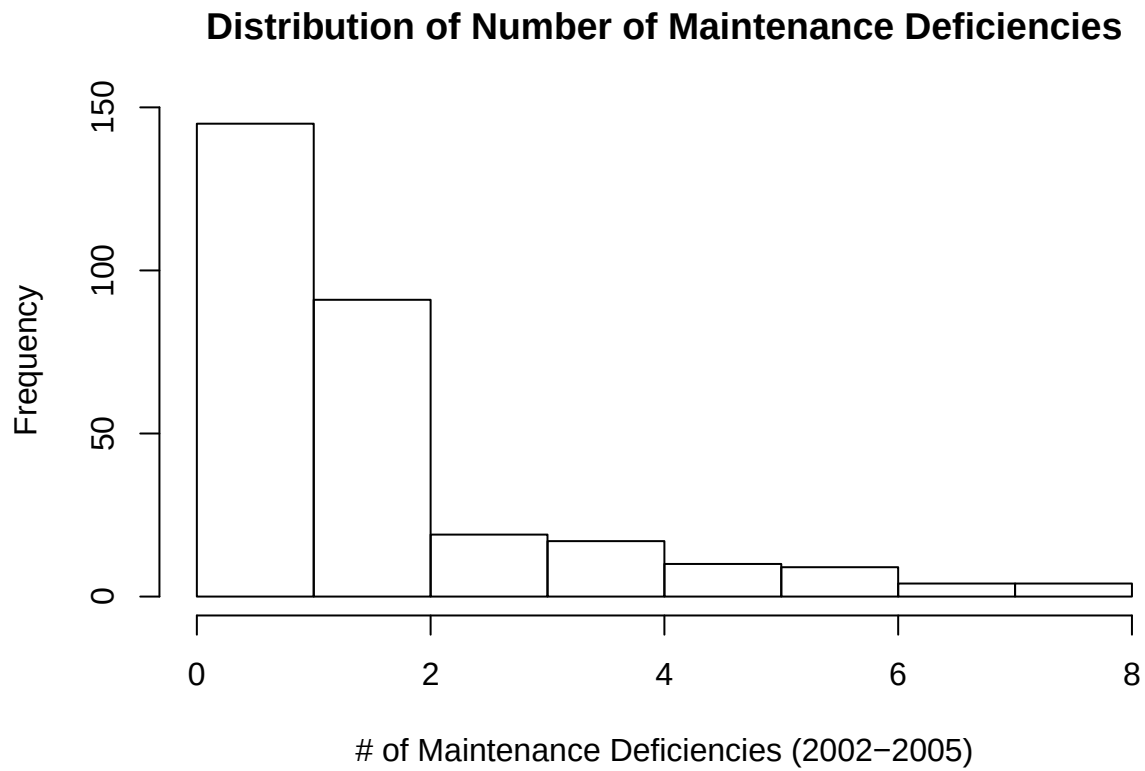


A numerical summary is supplemented as follows:

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	26.00	42.00	49.00	50.03	58.00	85.00

The distribution of *age* in NYC is approximately unimodal and right-skewed, with median and mean ages to both be around 50 years old.

Maintenance Deficiencies



A numerical summary is supplemented as follows:

```
summary(nyc$MaintenanceDef)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	0.00	1.00	2.00	1.98	2.00	8.00

The distribution of Maintenance Deficiencies appears to be exponentially decreasing, with a center of 2 maintenance deficiencies.

Moving Years

Distribution of NYC Residents' Moving Years



A numerical summary is supplemented as follows:

```
summary(nyc$NYCMove)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	1942	1973	1985	1983	1995	2004

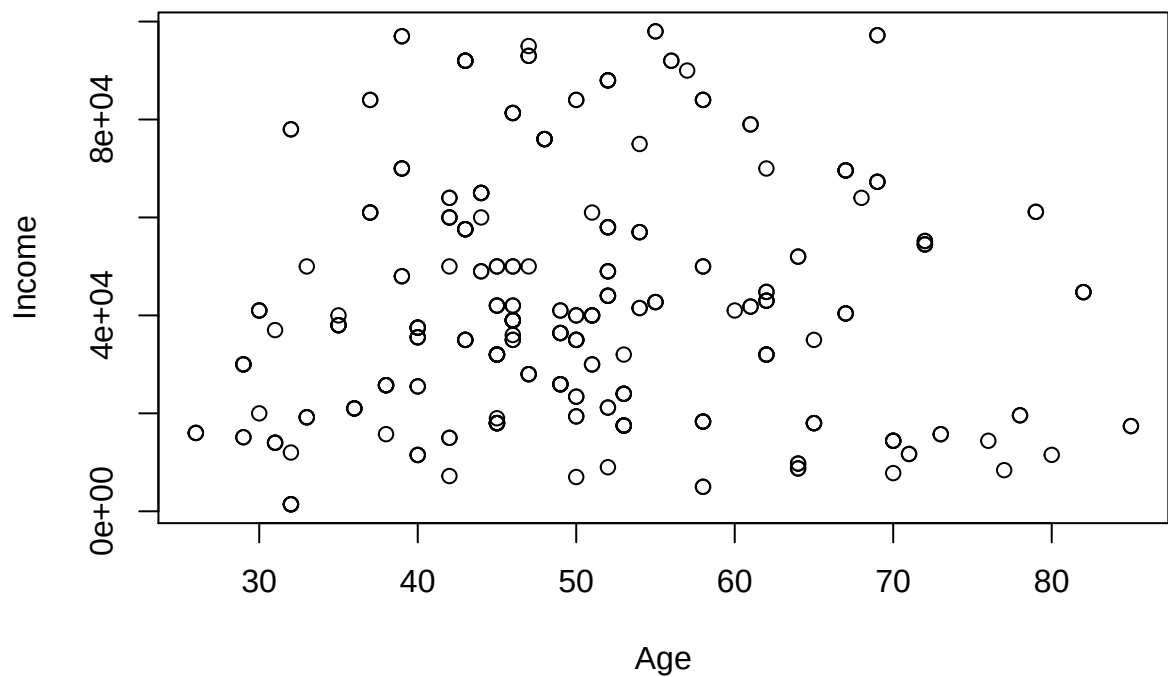
The distribution of *NYC move-in years* appears to be both unimodal and bimodal, with a left skew and an average move-in year of 1983. We notice a large temporary spike in residents moving in between 1975 and 1980, and this may be an interesting detail to note.

Bivariate Exploration

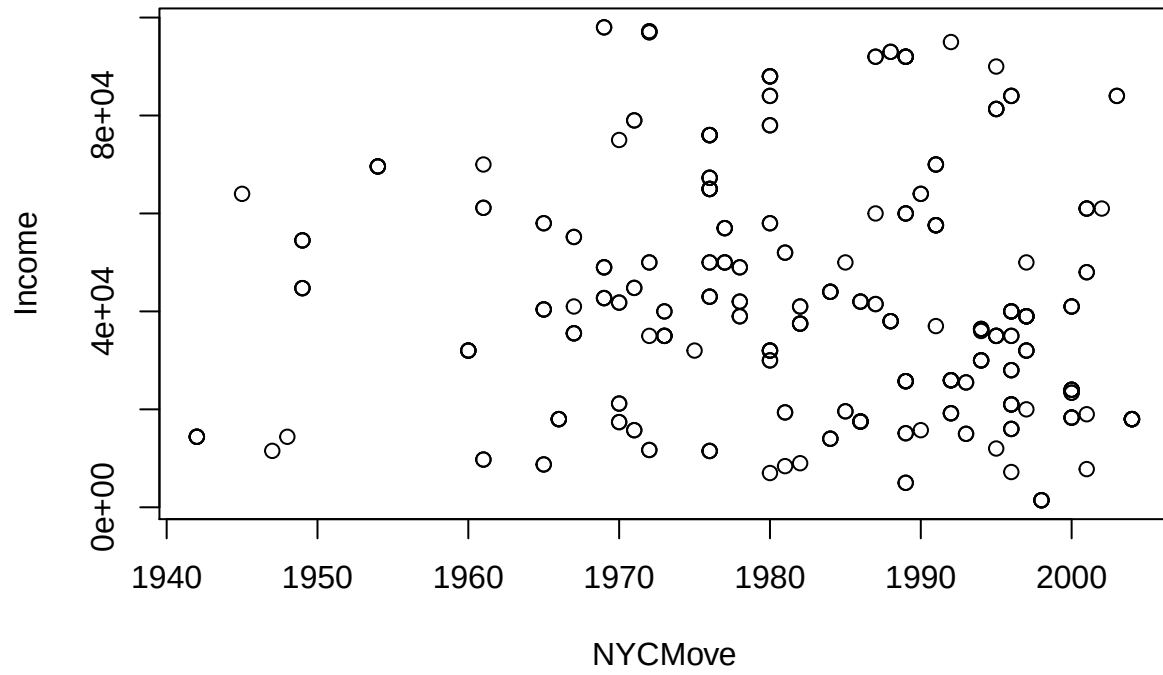
We understood from univariate exploration a general idea of the distributions of each variable. Now, we want to figure out if any of these variables are associated with *income*, our response variable.

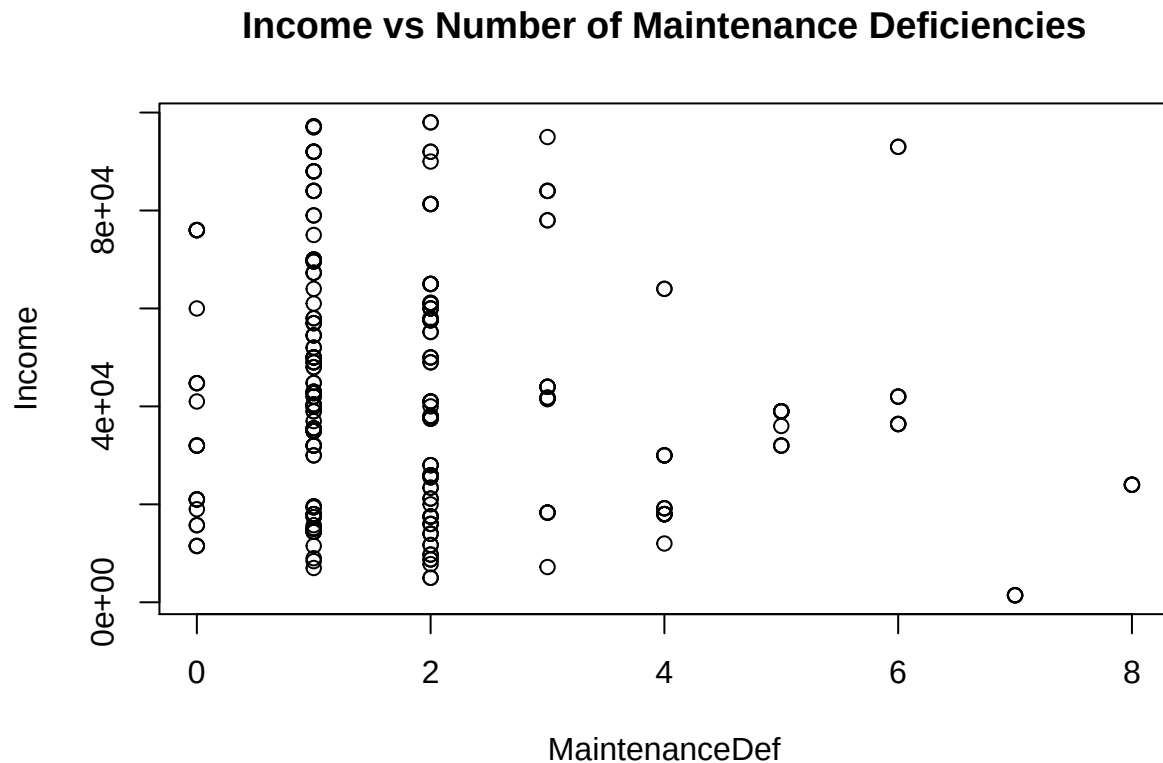
Three graphs are shown below, with x-axis' equal to each of the three predictor variables: *age*, *moving year*, and *maintenance deficiencies*.

Income vs Age of Respondents



Income vs Moving Years of Respondents





From our scatterplots, we do not notice any immediate correlations between *income* and any of the other variables. We will need to conduct more rigorous analysis to determine whether there is a correlation between *income* and the predictor variables.

Although none of the scatterplots show a strong correlation between two variables, we can argue:

- There is a **weak positive linear** correlation between age and income.
- There is a **weak negative linear** correlation between age and moving years.
- There is a **weak negative linear** correlation between age and maintenance deficiencies.

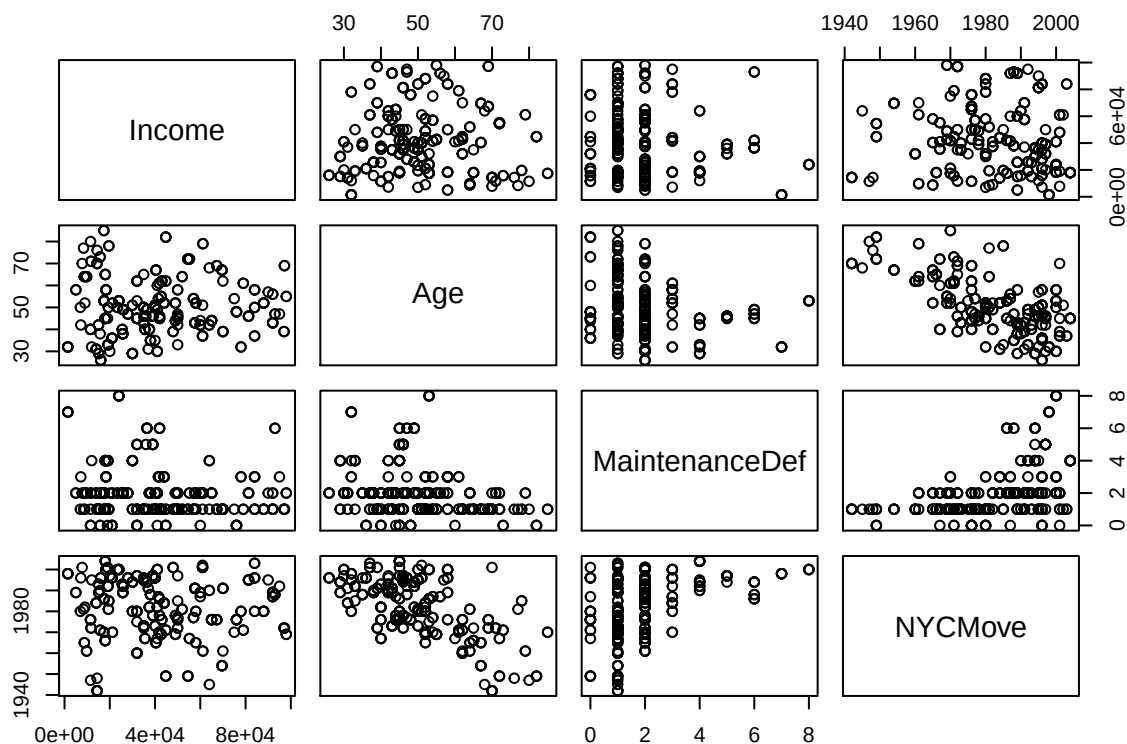
Modeling

From our exploratory data analysis, we got a sense of each of the variables' distributions and their relationships with our response variable, *income*.

We now want to formally determine if there is any relationship between the predictor and the response variables. In this section, we will build a linear regression model to predict patient satisfaction.

Before we build a linear model for predicting patient satisfaction, we first need to check for any unusually high multicollinearity between the response variables *age*, *moving years*, and *maintenance deficiencies*.

We will first construct a pair plot to get an idea of any strong correlation between pairs of explanatory variables.



From the pairs plot, we notice a strong negative linear relationship between *Age* and *NYCMove*, so we will need to check the level of multicollinearity between these two variables. We also notice a weak positive relationship with *MaintenanceDef* and *NYCMove*, and a weak negative relationship between *Age* and *MaintenanceDef*.

Let's explore the levels of linear multicollinearity using a variation inflation factors (vif) table, using a new model, *income_model*, to test multicollinearity:

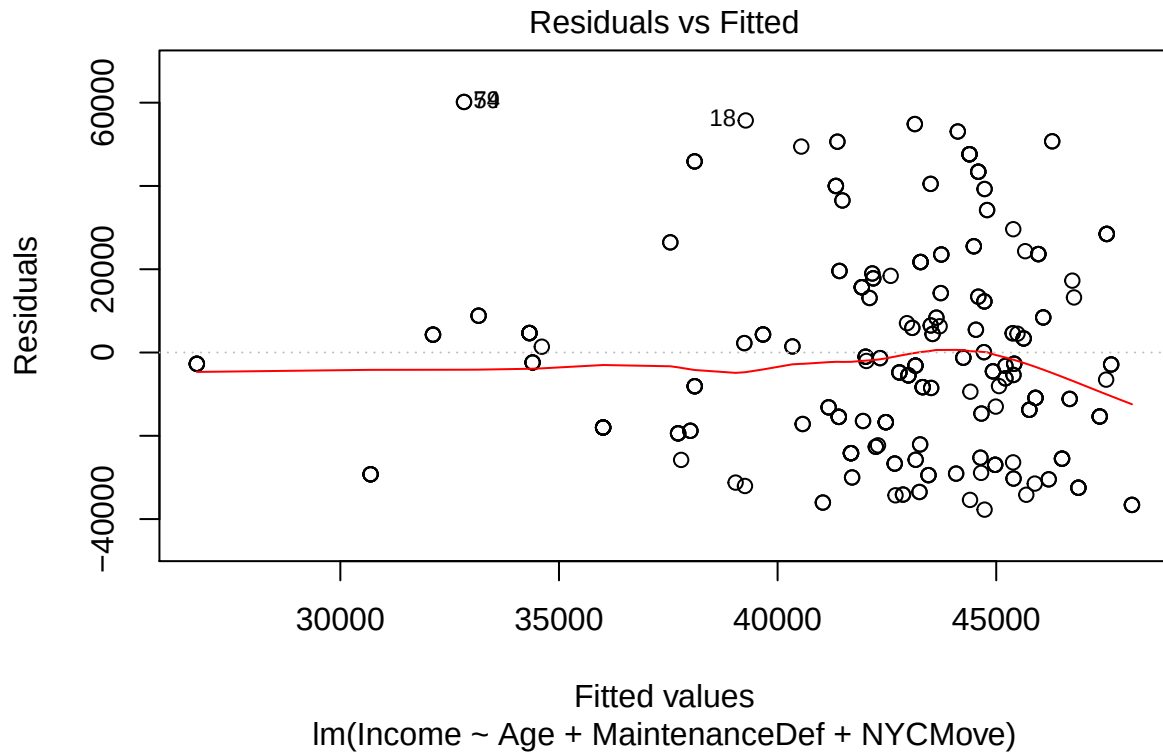
```
income_model <- lm(Income ~ Age + MaintenanceDef + NYCMove,
  data = nyc)
```

```
##           Age MaintenanceDef      NYCMove
##           1.687649           1.267728      1.999724
```

From the vif table, we can see that the levels of multicollinearity between the exploratory variables and with *Income* fall between 1 and 2. Because none of the values are greater than 2.5 (our threshold of dangerous multicollinearity), we do not have to be concerned about multicollinearity between each of the exploratory variables.

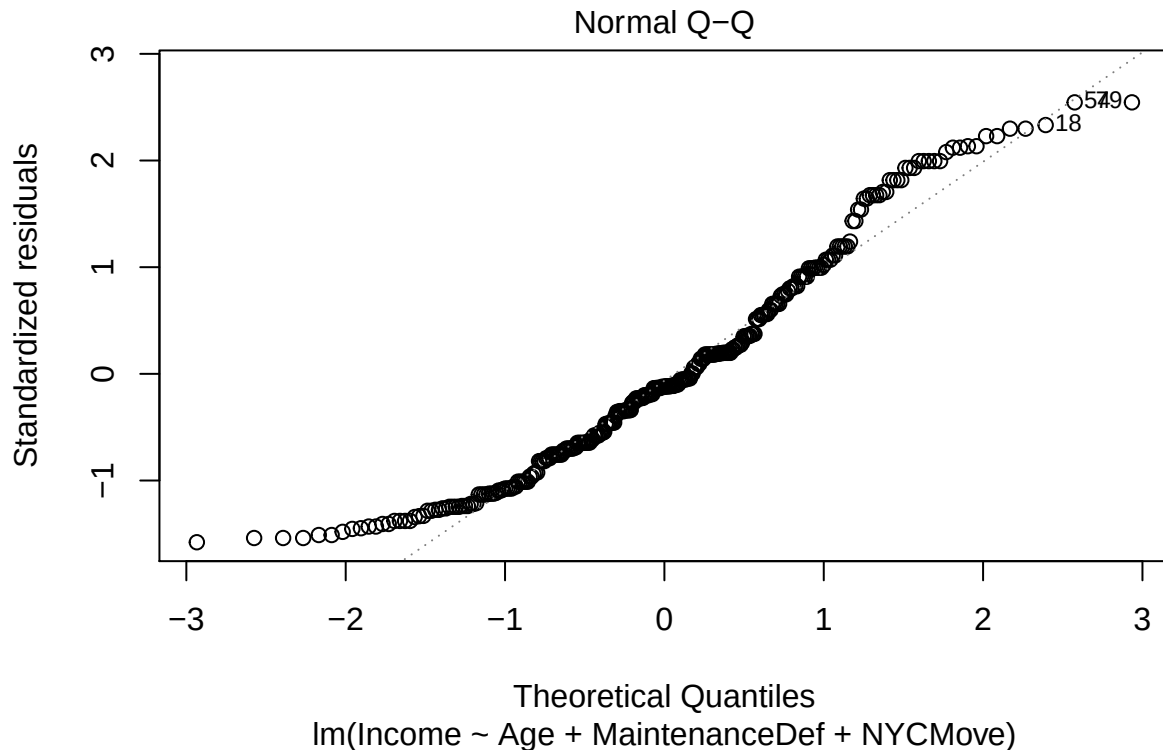
We can now move forward with our linear regression model, built with *Age*, *MaintenanceDef*, and *NYCMove* as exploratory variables for *Income*.

Below is a residual diagnostic plot using our built model, *income_model*:



We notice from our residual diagnostic plot that there is a reasonably constant and patternless vertical spread, with a consistent center around residual = 0.

Below is the qqplot representing this linear model:



From this qqplot, we notice that although the points deviate from the line on more extreme quantiles, they do not seem to invalidate the normality condition since most of the points fall reasonably near the line on the qqplot.

Let's produce a summary of the linear regression model we have made so far:

```
##
## Call:
## lm(formula = Income ~ Age + MaintenanceDef + NYCMove, data = nyc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -37734 -18010  -2878   14971   60171
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  237408.41  278939.01   0.851   0.3954
## Age          -71.98     144.97  -0.496   0.6199
## MaintenanceDef -2273.22    964.72  -2.356   0.0191 *
## NYCMove       -94.34     138.82  -0.680   0.4973
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 23960 on 295 degrees of freedom
## Multiple R-squared:  0.02981,    Adjusted R-squared:  0.01995
## F-statistic: 3.022 on 3 and 295 DF,  p-value: 0.03005
```

Our linear regression model seems to significantly fit the data, with a p-value of 0.03005. **However**, we

notice that the p-values of *Age* and *NYCMove* are both above 0.05, indicating that they are both insignificant predictors of *Income*.

To simplify the model, and to include only explanatory variables that will predict *Income*, we will create a **new model with just MaintenanceDef**.

Below is our new model, *income_model_adjusted*, that predicts *Income* with just *MaintenanceDef*

```
income_model_adjusted <- lm(Income ~ MaintenanceDef,
                             data = nyc)

##
## Call:
## lm(formula = Income ~ MaintenanceDef, data = nyc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -37727 -19004  -2727   15385   60831
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    47238.6     2184.7   21.622 < 2e-16 ***
## MaintenanceDef -2511.6       854.6   -2.939  0.00355 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 23900 on 297 degrees of freedom
## Multiple R-squared:  0.02826,    Adjusted R-squared:  0.02499
## F-statistic: 8.637 on 1 and 297 DF,  p-value: 0.003553
```

With this model, both the (*Intercept*) and *MaintenanceDef* are significant predictors of *Income*, with p-values less than 0.05.

Our linear regression model gets a p-value of 0.02499, indicating that the model significantly predicts *Income*. With our new model, we also obtain a higher Adjusted R-squared value, with a value of 0.02499 as opposed to the older model's value of 0.01995.

We are confident that the Number of Maintenance Deficiencies in a residence is associated with a resident's income.

To reiterate, our final model summary is shown below:

```
##
## Call:
## lm(formula = Income ~ MaintenanceDef, data = nyc)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -37727 -19004  -2727   15385   60831
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    47238.6     2184.7   21.622 < 2e-16 ***
## MaintenanceDef -2511.6       854.6   -2.939  0.00355 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 23900 on 297 degrees of freedom
```

```
## Multiple R-squared:  0.02826,    Adjusted R-squared:  0.02499
## F-statistic: 8.637 on 1 and 297 DF,  p-value: 0.003553
```

Prediction

We are interested in predicting the *Income* of a New York City resident whose age is 53 years old, moved to the city in 1987, and has 3 maintenance deficiencies from 2002-2005.

We will use our linear regression model to predict income. Our model states that:

$$\text{Income} = 47238.6 + (-2511.6) * \text{MaintenanceDef}$$

Plugging in our value of 3 for MaintenanceDef, we get:

```
47238.6 + (-2511.6) * 3
```

```
## [1] 39703.8
```

a predicted income of \$39,703, on average.

The predicted income of a New York City resident aged 53 years old, moved to NYC in 1987, and has had 3 maintenance deficiencies is \$39,703.

Discussion

In this report, we explored the relationship between a person's income with age, moving year, and their resident's number of maintenance deficiencies. We showed that a linear model could be made to predict income using all three variables. However, we also showed that not all three of the variables were significant predictors of a person's income.

To summarize, our analysis tells us that from our dataset, the amount of Maintenance Deficiencies is the only significant indicator of a NYC resident's Income.

Overall, using data to predict the income of residents in NYC will give us a sense of the requirements needed for anyone aspiring to live in New York City. Because the number of variables we analyzed were fairly limited, we hope that future surveyors and analysts will take more variables into account. By surveying people on more variables such as (but not limited to) daily expenses, location, and number of rooms in their residence, future data scientists will have the opportunity to provide better models for predicting the income of residents of New York City.